

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP pH	WQP 003	1.0	

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VERSION HISTORY			
Version	Date	Description	Approved By
1.0	5/10/23	Initial Release	TAC

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1. Title

Standard Operating Procedure for pH Determination of Seawater

2. Applicability

This SOP applies to the determination of *in situ* seawater pH profiles at shallow (<30 m) coral reef sites around Puerto Rico.

3. Purpose

- 3.1. Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to analyze seawater pH profiles at shallow coral reef sites around Puerto Rico.
- 3.2. Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and any other parties interested in coral reef seawater pH data following the Quality Assurance Project Plan.

4. Method

- 4.1. EPA Method 150.2 (Environmental Protection Agency).
- 4.2. Hydrolab HL4 Multiparameter Sonde is used to measure surface, bottom, and depth profiles of *in situ* seawater pH.

5. Health & Safety

- 5.1. Follow the safety indications for equipment and personnel suggested by the manufacturer.
- 5.2. When loosening removable parts of a manifold probe, face the probe away from your body and others, as pressure may build up within the instrument resulting in caps, sensors, or other parts that can shoot out and cause injury.

6. General Precautions

- 6.1. Always maintain two points of contact when deploying the HL4 sonde on the boat to avoid losing the instrument overboard.
- 6.2. Ensure the boat engines are idle and the captain approves of the sonde being deployed overboard before deploying the sonde to ensure that the instrument cables do not become entangled with the boat propeller.
- 6.3. During maintenance or replacement of the multiprobe battery, place it horizontally on a planar surface before removing the battery cap to avoid water leaks into the multiprobe battery compartment during maintenance or replacement.
- 6.4. In case water leaks into the battery compartment, remove the batteries, pour the water out, thoroughly dry the compartment, and install new batteries.
- 6.5. If there are problems with the software, call the manufacturer (800-949-3766) or email natechsupport@otthydromet.com.
- 6.6. HydroLab HL4 Sonde requires that sensors are calibrated in the following order:
 - 6.6.1. Temperature (by manufacturer only) → conductivity → LDO → pH
- 6.7. Do not allow the sensors on the HydroLab HL4 Sonde to dry.

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6.8. For storage after deployment:

- 6.8.1. Rinse the handheld surveyor, the sonde sensors and the sensor guard with tap water,
- 6.8.2. Rinse the storage cup with tap water,
- 6.8.3. Remove the sensor guard and replace it with the storage cup while it is still wet to allow the sensors to remain humid but **not submerged**.

7. Training

- 7.1 Personnel shall be trained to calibrate and operate the multiparameter HL4 sonde before use
- 7.2 Personnel shall receive training on how to complete the corresponding forms for this SOP.

8. Required Materials

- 8.1 Hydrolab HL4 Multiparameter Sonde
- 8.2 Hydrolab Handheld Surveyor
- 8.3 Handheld Surveyor USB cable
- 8.4 Laptop computer with HydroLab software installed
- 8.5 Mild detergent to clean the sensor
- 8.6 Calibration standards pH 7, 9, and 10
- 8.7 Warm water to clean the sensor
- 8.8 Soft toothbrush to clean the sensor
- 8.9 Extra “D” battery for replacement
- 8.10 Calibration Form (Appendix I)
- 8.11 Field Form (Appendix F)

9. Procedure

9.1 Calibration

Calibration Frequency: The Hydrolab HL4 Multiparameter Sonde (OTT HydroMet) should be calibrated following manufacturers recommendations (~3 months) to ensure accuracy and control for any potential sensor drift throughout the sampling event.

- 9.1.1 Place the HL4 Sonde and handheld Surveyor® near the sink to start the calibration process.
- 9.1.2 Connect the sonde to the handheld Surveyor® HL and press the **power button** on the handheld.
- 9.1.3 Press the **menu button** → **RIGHT arrow** until you read “Sonde_HL4” on the top of the display window.
- 9.1.4 Select “Calibration” → select “pH” → select “pH 3-point Calibration”.
- 9.1.5 Always calibrate the pH sensor using a 3-point calibration.
- 9.1.6 Fill the storage cup 1/3 of its capacity with room temperature distilled water → attach the storage cup to the sonde → shake vigorously for at least 6 seconds → discard the water in the storage cup.
 - 9.1.6.1 Repeat three times total.

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- 9.1.7 Fill the storage cup with pH standard 7 to the black rubber cover → attach the storage cup to the sonde → shake vigorously for at least 6 seconds → discard the water from the storage cup.
 - 9.1.7.1 Repeat three times total.
- 9.1.8 Fill the storage cup 3/4 of its capacity with pH 7.0 standard or enough to submerge all sensors when the sonde is held upside down (sensors facing downwards).
 - 9.1.8.1 Before accepting calibration, tilt the HL4 sonde in different directions and gently tap the storage cup with the standard solution to remove any bubbles stuck on the sensors.
 - 9.1.8.2 Allow values to stabilize for 1 minute → write down “7.00” on the provided text-box and press **OK button** → wait for the stabilization bar to read “Stable” → press **OK button** to continue → discard pH standard from the storage cup in the appropriate container.
- 9.1.9 Repeat the room temperature distilled water rinse on step 9.1.5.
- 9.1.10 Repeat steps 9.1.6. - 9.1.8. using pH 9 and pH 10 standards.
- 9.1.11 Complete software calibration form:
 - 9.1.11.1 User name
 - 9.1.11.2 Log Notes - Include the names of all people involved in the calibration.
 - 9.1.11.3 Select “Keep Calibration Date”.
 - 9.1.11.4 Record calibration results in Form Appendix I.
- 9.1.12 Repeat distilled water rinse on step 9.1.5. and prepare for storage.

9.2 Check Calibration

A calibration check will be performed at the beginning of each sampling day.

- 9.2.1 Place the HL4 Sonde and handheld Surveyor® near the sink before starting the calibration check.
- 9.2.2 Connect the sonde to the handheld Surveyor® HL and press the **power button** on the handheld.
- 9.2.3 Press the **menu button** → **RIGHT arrow** until you read “Sonde_HL4” on the top of the display window.
- 9.2.4 Select “*Calibration*” → select “*pH*” → select “*Check Calibration*”
- 9.2.5 Follow instructions on the handheld display.
- 9.2.6 Rinse the storage cup and sensors 3 times using regular tap water → place the tap water-humid storage cup on the sonde until it is ready to be deployed.
- 9.2.7 Disconnect the sonde from the calibration cable.

9.3 Prepare the sonde for deployment.

- 9.3.1 Connect the sonde to the handheld Surveyor® HL using the deployment cable and press the power button on the handheld.
- 9.3.2 After checking calibration for each sensor, attach the tap water-humid storage cup to the sonde until arrival to the sampling site.
- 9.3.3 Once at the sampling site, replace the storage cup with the sensor guard for deployment.

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9.3.4 At the beginning and end of each 42 site sampling cycle, perform an additional check using Dickson Lab Certified Reference Material Tris Solution to calculate pH on the total scale.

9.3.4.1 Analyze the Tris Solution as a single measurement and record the measured pH_{NBS} reported by the sonde and the reported pH_T for the Certified Reference Material.

9.4 Sample Collection

9.4.1 While the HL4 sonde is connected to the handheld and deployment cable, press the **power button** → **RIGHT arrow** until you read “Sonde_HL4”.

9.4.1.1 Select “Overview” and make sure all sensors have started properly (there should be a check mark to the right of each parameter (e.g. pH sensor ✓).

9.4.1.1.1 Once checked, press the **return button** ↵.

9.4.2 Without placing the HL4 sonde into the water, and while on the “Sonde_HL4” menu, select “Real-time” → press **OK button** to start data save.

9.4.2.1 Press the **DOWN arrow button** once and then the **UP arrow button** once to select “New Log” → press **OK button**

9.4.2.1.1 Replicate casts should be saved as “SiteCode_R2” instead of the standard log name. **Otherwise, leave the default log name.**

9.4.2.2 Select “Location” and select the corresponding “Site Code” → press **OK button**.

9.4.2.3 Press the **RIGHT arrow button** to save the settings → press **OK button** to begin data recording (you should see the number of “Measurements” on the lower left corner of the display window steadily increasing).

9.4.3 Gently lower the HL4 Sonde ~0.5 m into the water → using Sampling Log (Appendix F) record the desired parameters and their respective depths, **leaving surface temperature for last** to allow for stabilization of the sensor.

9.4.4 Lower the sonde at a slow and steady rate until it gets ~ 0.5m from the bottom with caution not to touch the bottom and stir sediment up.

9.4.5 Record the bottom pH (pH units) and corresponding depth (m) reading on the Sampling Log (Appendix F).

9.4.6 Pull the sonde back up into the boat.

9.4.7 Press **OK button** to stop recording the cast → press **return button** ↵ → turn off the handheld surveyor.

9.4.8 If other sites are to be sampled, place the Hydrolab sonde on a bucket that holds clean tap water until arrival at the next site.

9.4.8.1 If no other sites are to be sampled, follow **step 6.8 for storage after deployment**.

9.5 Transfer data from handheld Surveyor to PC

9.5.1 Connect the surveyor to the PC using the manufacturer's USB cable and press the power button on the handheld.

9.5.2 Open the *Hydrolab Operating Software* → select the handheld surveyor from the available devices list → click on “Overview” on the navigation bar.

9.5.3 On the lower-right of the display, select the log files you wish to transfer and click “Transfer...”. → select save folder → click “Save” (these will be in Hydrolab File Format).

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9.5.4 Open the Hydrolab File Format saved files → click “Export...” → click **OK** → rename the file as: *SiteCode_Date of collection_default log name* (e.g., BOYA20_12MAY23_Log_12345_6789) → select format type (e.g., CSV, XLSX) → click “Save”.

10. Documentation

- 10.1 Field data will be recorded in Field Form (Appendix F).
- 10.2 Calibration data will be recorded in Calibration Form (Appendix I).
- 10.3 Field and Calibration forms shall be kept in the designated file cabinet at the CCRI office.
- 10.4 Data contained in hard copy forms shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed
- 10.5 All spreadsheets and depth profile logs should be uploaded to the internal CCRI Water Quality Database
- 10.6 All data will be electronically archived and backed up online.

11. Cleaning and Maintenance

- 11.1 After each sampling event, rinse the sensors, the storage cup, and the sensor guard thoroughly with clean warm water.
- 11.2 After each sampling event, inspect sensors carefully for physical damage.
 - 11.2.1 Pay particular attention to the dissolved oxygen probe membrane. Look for wrinkling or other physical damage.
- 11.3 In case of unwanted material deposits over the sensors, gently clean sensors with a soft brush and mild detergent. Do not use dishwashing liquid or detergents with skin moisturizer as these can create films over the probes.
- 11.4 Fill the storage cup with tap water for long term storage.
 - 11.4.1 Do not fill the storage cup with distilled water as this may damage the probes.
- 11.5 Replace sonde “D” battery as necessary.
 - 11.5.1 The software will indicate when the battery charge is low.
- 11.6 Perform a monthly preventive maintenance
 - 11.6.1 Carefully inspect sensors, O-rings, connective pins, and physical integrity of the sonde in general.
 - 11.6.2 Lubricate or replace O-rings, if necessary.
 - 11.6.2.1 Refer to HL Series Sensors DOC026.97.80349 document for instructions.
- 11.7 Follow manufacturer indications for maintenance.
 - 11.7.1 Refer to “HL Series Sonde DOC026.97.80348” document for instructions.

12. Troubleshooting

- 12.1 In case of reading errors, check the sensor path for bubbles, grease, or other contamination.
- 12.2 In case of unstable readings or calibration, make sure standards and distilled water used for rinsing are at room temperature.
 - 12.2.1 If problems persist, contact the manufacturer to replace the sensor.

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13. QA/QC

- 13.1 As per QAPP Table 4, seawater pH should be measured with an accuracy of ± 0.2 ($\pm 10\%$) and a precision of $< 10\%$.
- 13.2 In accordance with the QAPP, 10% of all Hydrolab casts will be repeated to ensure precision of replicate surface and bottom measurements is $< 20\%$.
- 13.3 All field samples should adhere to typical seasonal pH values observed in coastal waters of Puerto Rico.
- 13.4 Any seemingly erroneous measurements will be flagged for further review in accordance with the QAPP.

14. References

- 14.1 Annual Book of ASTM Standards, Part 31, "Water" Standard 1293-78, Method D, p. 226 (1981)
- 14.2 Hydrolab (2014). Hydrolab Operating Software. DOC026.53.80346I
- 14.3 Hydrolab (2017). HL Series Sonde User Manual. DOC026.97.80348
- 14.4 Hydrolab (2019). HL Series Sensors User Manual DOC026.97.80349